

29. Let $\{x_n\}$ be the sequence of population values defined recursively by $x_1 = 60$, and for $n \geq 1$, x_{n+1} is given by the Beverton–Holt model with $R = 10$ and $K = 300$.
- List the first four terms of the sequence $\{x_n\}$.
 - If $0 < x_n < 300$, show that $0 < x_{n+1} < 300$. Conclude that $0 < x_n < 300$ for $n \geq 1$.
 - Show that $\{x_n\}$ is increasing.
 - Show that $\{x_n\}$ converges and find its limit L .
30. Let $\{x_n\}$ be a sequence of population values defined recursively by the Beverton–Holt model for which $x_1 > K$. Assume that the constants R and K satisfy $R > 1$ and $K > 0$.
- If $x_n > K$, show that $x_{n+1} > K$. Conclude that $x_n > K$ for all $n \geq 1$.
 - Show that $\{x_n\}$ is decreasing.
 - Show that $\{x_n\}$ converges and find its limit L .
31. The goal of this exercise is to establish Formula (5), namely,

$$\lim_{n \rightarrow +\infty} \frac{x^n}{n!} = 0$$

Let $a_n = |x|^n/n!$ and observe that the case where $x = 0$ is obvious, so we will focus on the case where $x \neq 0$.

- Show that
- $$a_{n+1} = \frac{|x|}{n+1} a_n$$
- Show that the sequence $\{a_n\}$ is eventually strictly decreasing.
 - Show that the sequence $\{a_n\}$ converges.
32. (a) Compare appropriate areas in the accompanying figure to deduce the following inequalities for $n \geq 2$:

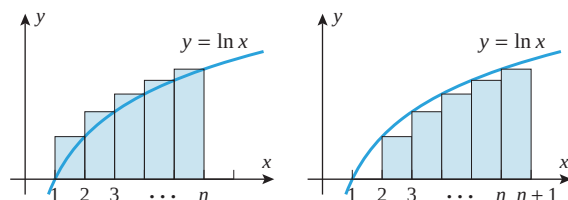
$$\int_1^n \ln x \, dx < \ln n! < \int_1^{n+1} \ln x \, dx$$

- Use the result in part (a) to show that

$$\frac{n^n}{e^{n-1}} < n! < \frac{(n+1)^{n+1}}{e^n}, \quad n > 1$$

- Use the Squeezing Theorem for Sequences (Theorem 9.1.5) and the result in part (b) to show that

$$\lim_{n \rightarrow +\infty} \frac{\sqrt[n]{n!}}{n} = \frac{1}{e}$$



▲ Figure Ex-32

33. Use the left inequality in Exercise 32(b) to show that

$$\lim_{n \rightarrow +\infty} \sqrt[n]{n!} = +\infty$$

34. **Writing** Give an example of an increasing sequence that is not eventually strictly increasing. What can you conclude about the terms of any such sequence? Explain.
35. **Writing** Discuss the appropriate use of “eventually” for various properties of sequences. For example, which is a useful expression: “eventually bounded” or “eventually monotone”?

✓ QUICK CHECK ANSWERS 9.2

1. I; D; N; I; N 2. N; M; S 3. 1; increasing 4. 8; eventually; increasing

9.3 INFINITE SERIES

The purpose of this section is to discuss sums that contain infinitely many terms. The most familiar examples of such sums occur in the decimal representations of real numbers. For example, when we write $\frac{1}{3}$ in the decimal form $\frac{1}{3} = 0.3333 \dots$, we mean

$$\frac{1}{3} = 0.3 + 0.03 + 0.003 + 0.0003 + \dots$$

which suggests that the decimal representation of $\frac{1}{3}$ can be viewed as a sum of infinitely many real numbers.

■ SUMS OF INFINITE SERIES

Our first objective is to define what is meant by the “sum” of infinitely many real numbers. We begin with some terminology.

9.3.1 DEFINITION An *infinite series* is an expression that can be written in the form

$$\sum_{k=1}^{\infty} u_k = u_1 + u_2 + u_3 + \cdots + u_k + \cdots$$

The numbers $u_1, u_2, u_3, \dots$ are called the **terms** of the series.

Since it is impossible to add infinitely many numbers together directly, sums of infinite series are defined and computed by an indirect limiting process. To motivate the basic idea, consider the decimal

$$0.3333 \dots \quad (1)$$

This can be viewed as the infinite series

$$0.3 + 0.03 + 0.003 + 0.0003 + \cdots$$

or, equivalently,

$$\frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} + \frac{3}{10^4} + \cdots \quad (2)$$

Since (1) is the decimal expansion of $\frac{1}{3}$, any reasonable definition for the sum of an infinite series should yield $\frac{1}{3}$ for the sum of (2). To obtain such a definition, consider the following sequence of (finite) sums:

$$\begin{aligned} s_1 &= \frac{3}{10} = 0.3 \\ s_2 &= \frac{3}{10} + \frac{3}{10^2} = 0.33 \\ s_3 &= \frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} = 0.333 \\ s_4 &= \frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} + \frac{3}{10^4} = 0.3333 \\ &\vdots \end{aligned}$$

The sequence of numbers $s_1, s_2, s_3, s_4, \dots$ (Figure 9.3.1) can be viewed as a succession of approximations to the “sum” of the infinite series, which we want to be $\frac{1}{3}$. As we progress through the sequence, more and more terms of the infinite series are used, and the approximations get better and better, suggesting that the desired sum of $\frac{1}{3}$ might be the *limit* of this sequence of approximations. To see that this is so, we must calculate the limit of the general term in the sequence of approximations, namely,

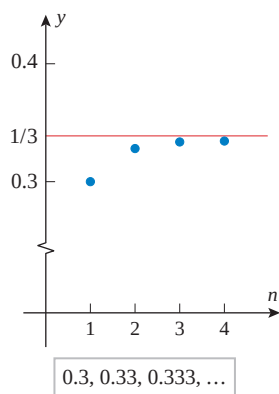
$$s_n = \frac{3}{10} + \frac{3}{10^2} + \cdots + \frac{3}{10^n} \quad (3)$$

The problem of calculating

$$\lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} \left(\frac{3}{10} + \frac{3}{10^2} + \cdots + \frac{3}{10^n} \right)$$

is complicated by the fact that both the last term and the number of terms in the sum change with n . It is best to rewrite such limits in a closed form in which the number of terms does not vary, if possible. (See the discussion of closed form and open form following Example 2 in Section 5.4.) To do this, we multiply both sides of (3) by $\frac{1}{10}$ to obtain

$$\frac{1}{10} s_n = \frac{3}{10^2} + \frac{3}{10^3} + \cdots + \frac{3}{10^n} + \frac{3}{10^{n+1}} \quad (4)$$



▲ Figure 9.3.1

and then subtract (4) from (3) to obtain

$$\begin{aligned}s_n - \frac{1}{10}s_n &= \frac{3}{10} - \frac{3}{10^{n+1}} \\ \frac{9}{10}s_n &= \frac{3}{10} \left(1 - \frac{1}{10^n}\right) \\ s_n &= \frac{1}{3} \left(1 - \frac{1}{10^n}\right)\end{aligned}$$

Since $1/10^n \rightarrow 0$ as $n \rightarrow +\infty$, it follows that

$$\lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} \frac{1}{3} \left(1 - \frac{1}{10^n}\right) = \frac{1}{3}$$

which we denote by writing

$$\frac{1}{3} = \frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} + \cdots + \frac{3}{10^n} + \cdots$$

Motivated by the preceding example, we are now ready to define the general concept of the “sum” of an infinite series

$$u_1 + u_2 + u_3 + \cdots + u_k + \cdots$$

We begin with some terminology: Let s_n denote the sum of the initial terms of the series, up to and including the term with index n . Thus,

$$\begin{aligned}s_1 &= u_1 \\ s_2 &= u_1 + u_2 \\ s_3 &= u_1 + u_2 + u_3 \\ &\vdots \\ s_n &= u_1 + u_2 + u_3 + \cdots + u_n = \sum_{k=1}^n u_k\end{aligned}$$

The number s_n is called the ***n*th partial sum** of the series and the sequence $\{s_n\}_{n=1}^{+\infty}$ is called the ***sequence of partial sums***.

As n increases, the partial sum $s_n = u_1 + u_2 + \cdots + u_n$ includes more and more terms of the series. Thus, if s_n tends toward a limit as $n \rightarrow +\infty$, it is reasonable to view this limit as the sum of *all* the terms in the series. This suggests the following definition.

9.3.2 DEFINITION Let $\{s_n\}$ be the sequence of partial sums of the series

$$u_1 + u_2 + u_3 + \cdots + u_k + \cdots$$

If the sequence $\{s_n\}$ converges to a limit S , then the series is said to ***converge*** to S , and S is called the ***sum*** of the series. We denote this by writing

$$S = \sum_{k=1}^{\infty} u_k$$

If the sequence of partial sums diverges, then the series is said to ***diverge***. A divergent series has no sum.

WARNING

In everyday language the words “sequence” and “series” are often used interchangeably. However, in mathematics there is a distinction between these two words—a sequence is a *succession* whereas a series is a *sum*. It is essential that you keep this distinction in mind.

► **Example 1** Determine whether the series

$$1 - 1 + 1 - 1 + 1 - 1 + \cdots$$

converges or diverges. If it converges, find the sum.

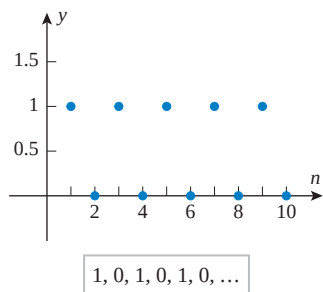
Solution. It is tempting to conclude that the sum of the series is zero by arguing that the positive and negative terms cancel one another. However, this is *not correct*; the problem is that algebraic operations that hold for finite sums do not carry over to infinite series in all cases. Later, we will discuss conditions under which familiar algebraic operations can be applied to infinite series, but for this example we turn directly to Definition 9.3.2. The partial sums are

$$\begin{aligned}s_1 &= 1 \\s_2 &= 1 - 1 = 0 \\s_3 &= 1 - 1 + 1 = 1 \\s_4 &= 1 - 1 + 1 - 1 = 0\end{aligned}$$

and so forth. Thus, the sequence of partial sums is

$$1, 0, 1, 0, 1, 0, \dots$$

(Figure 9.3.2). Since this is a divergent sequence, the given series diverges and consequently has no sum. ◀



▲ Figure 9.3.2

GEOMETRIC SERIES

In many important series, each term is obtained by multiplying the preceding term by some fixed constant. Thus, if the initial term of the series is a and each term is obtained by multiplying the preceding term by r , then the series has the form

$$\sum_{k=0}^{\infty} ar^k = a + ar + ar^2 + ar^3 + \cdots + ar^k + \cdots \quad (a \neq 0) \quad (5)$$

Such series are called **geometric series**, and the number r is called the **ratio** for the series. Here are some examples:

$$1 + 2 + 4 + 8 + \cdots + 2^k + \cdots \quad a = 1, r = 2$$

$$\frac{3}{10} + \frac{3}{10^2} + \frac{3}{10^3} + \cdots + \frac{3}{10^k} + \cdots \quad a = \frac{3}{10}, r = \frac{1}{10}$$

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{8} - \frac{1}{16} + \cdots + (-1)^{k+1} \frac{1}{2^k} + \cdots \quad a = \frac{1}{2}, r = -\frac{1}{2}$$

$$1 + 1 + 1 + \cdots + 1 + \cdots \quad a = 1, r = 1$$

$$1 - 1 + 1 - 1 + \cdots + (-1)^{k+1} + \cdots \quad a = 1, r = -1$$

$$1 + x + x^2 + x^3 + \cdots + x^k + \cdots \quad a = 1, r = x$$

Sometimes it is desirable to start the index of summation of an infinite series at $k = 0$ rather than $k = 1$, in which case we would call u_0 the *zeroth term* and $s_0 = u_0$ the *zeroth partial sum*. One can prove that changing the starting value for the index of summation of an infinite series has no effect on the convergence, the divergence, or the sum. If we had started the index at $k = 1$ in (5), then the series would be expressed as

$$\sum_{k=1}^{\infty} ar^{k-1}$$

Since this expression is more complicated than (5), we started the index at $k = 0$.

The following theorem is the fundamental result on convergence of geometric series.

9.3.3 THEOREM A geometric series

$$\sum_{k=0}^{\infty} ar^k = a + ar + ar^2 + \cdots + ar^k + \cdots \quad (a \neq 0)$$

converges if $|r| < 1$ and diverges if $|r| \geq 1$. If the series converges, then the sum is

$$\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$$

PROOF Let us treat the case $|r| = 1$ first. If $r = 1$, then the series is

$$a + a + a + a + \cdots$$

so the n th partial sum is $s_n = (n + 1)a$ and

$$\lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} (n + 1)a = \pm\infty$$

(the sign depending on whether a is positive or negative). This proves divergence. If $r = -1$, the series is

$$a - a + a - a + \cdots$$

so the sequence of partial sums is

$$a, 0, a, 0, a, 0, \dots$$

which diverges.

Now let us consider the case where $|r| \neq 1$. The n th partial sum of the series is

$$s_n = a + ar + ar^2 + \cdots + ar^n \quad (6)$$

Multiplying both sides of (6) by r yields

$$rs_n = ar + ar^2 + \cdots + ar^n + ar^{n+1} \quad (7)$$

and subtracting (7) from (6) gives

$$s_n - rs_n = a - ar^{n+1}$$

or

$$(1 - r)s_n = a - ar^{n+1} \quad (8)$$

Since $r \neq 1$ in the case we are considering, this can be rewritten as

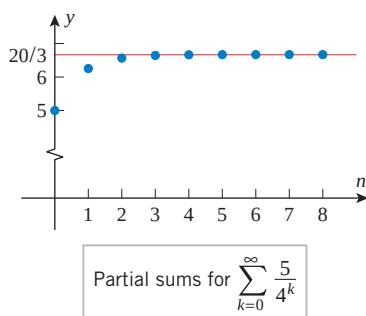
$$s_n = \frac{a - ar^{n+1}}{1 - r} = \frac{a}{1 - r}(1 - r^{n+1}) \quad (9)$$

Note that (6) is an open form for s_n , while (9) is a closed form for s_n . In general, one needs a closed form to calculate the limit.

If $|r| < 1$, then r^{n+1} goes to 0 as $n \rightarrow +\infty$ (can you see why?), so $\{s_n\}$ converges. From (9)

$$\lim_{n \rightarrow +\infty} s_n = \frac{a}{1 - r}$$

If $|r| > 1$, then either $r > 1$ or $r < -1$. In the case $r > 1$, r^{n+1} increases without bound as $n \rightarrow +\infty$, and in the case $r < -1$, r^{n+1} oscillates between positive and negative values that grow in magnitude, so $\{s_n\}$ diverges in both cases. ■



▲ Figure 9.3.3

► **Example 2** In each part, determine whether the series converges, and if so find its sum.

$$(a) \sum_{k=0}^{\infty} \frac{5}{4^k} \quad (b) \sum_{k=1}^{\infty} 3^{2k} 5^{1-k}$$

Solution (a). This is a geometric series with $a = 5$ and $r = \frac{1}{4}$. Since $|r| = \frac{1}{4} < 1$, the series converges and the sum is

$$\frac{a}{1 - r} = \frac{5}{1 - \frac{1}{4}} = \frac{20}{3}$$

(Figure 9.3.3).

Solution (b). This is a geometric series in concealed form, since we can rewrite it as

$$\sum_{k=1}^{\infty} 3^{2k} 5^{1-k} = \sum_{k=1}^{\infty} \frac{9^k}{5^{k-1}} = \sum_{k=1}^{\infty} 9 \left(\frac{9}{5}\right)^{k-1}$$

Since $r = \frac{9}{5} > 1$, the series diverges. ◀

TECHNOLOGY MASTERY

Computer algebra systems have commands for finding sums of convergent series. If you have a CAS, use it to compute the sums in Examples 2 and 3.

► **Example 3** Find the rational number represented by the repeating decimal

$$0.784784784 \dots$$

Solution. We can write

$$0.784784784 \dots = 0.784 + 0.000784 + 0.000000784 + \dots$$

so the given decimal is the sum of a geometric series with $a = 0.784$ and $r = 0.001$. Thus,

$$0.784784784 \dots = \frac{a}{1-r} = \frac{0.784}{1-0.001} = \frac{0.784}{0.999} = \frac{784}{999} \quad \blacktriangleleft$$

► **Example 4** In each part, find all values of x for which the series converges, and find the sum of the series for those values of x .

$$(a) \sum_{k=0}^{\infty} x^k \quad (b) \quad 3 - \frac{3x}{2} + \frac{3x^2}{4} - \frac{3x^3}{8} + \dots + \frac{3(-1)^k}{2^k} x^k + \dots$$

Solution (a). The expanded form of the series is

$$\sum_{k=0}^{\infty} x^k = 1 + x + x^2 + \dots + x^k + \dots$$

The series is a geometric series with $a = 1$ and $r = x$, so it converges if $|x| < 1$ and diverges otherwise. When the series converges its sum is

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$$

Solution (b). This is a geometric series with $a = 3$ and $r = -x/2$. It converges if $|-x/2| < 1$, or equivalently, when $|x| < 2$. When the series converges its sum is

$$\sum_{k=0}^{\infty} 3 \left(-\frac{x}{2}\right)^k = \frac{3}{1 - \left(-\frac{x}{2}\right)} = \frac{6}{2+x} \quad \blacktriangleleft$$

TELESCOPING SUMS

► **Example 5** Determine whether the series

$$\sum_{k=1}^{\infty} \frac{1}{k(k+1)} = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \frac{1}{4 \cdot 5} + \dots$$

converges or diverges. If it converges, find the sum.

Solution. The n th partial sum of the series is

$$s_n = \sum_{k=1}^n \frac{1}{k(k+1)} = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{n(n+1)}$$

We will begin by rewriting s_n in closed form. This can be accomplished by using the method of partial fractions to obtain (verify)

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$$

The sum in (10) is an example of a **telescoping sum**. The name is derived from the fact that in simplifying the sum, one term in each parenthetical expression cancels one term in the next parenthetical expression, until the entire sum collapses (like a folding telescope) into just two terms.

from which we obtain the sum

$$\begin{aligned} s_n &= \sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right) \\ &= \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1} \right) \\ &= 1 + \left(-\frac{1}{2} + \frac{1}{2} \right) + \left(-\frac{1}{3} + \frac{1}{3} \right) + \cdots + \left(-\frac{1}{n} + \frac{1}{n} \right) - \frac{1}{n+1} \\ &= 1 - \frac{1}{n+1} \end{aligned} \quad (10)$$

Thus,

$$\sum_{k=1}^{\infty} \frac{1}{k(k+1)} = \lim_{n \rightarrow +\infty} s_n = \lim_{n \rightarrow +\infty} \left(1 - \frac{1}{n+1} \right) = 1 \quad \blacktriangleleft$$

HARMONIC SERIES

One of the most important of all diverging series is the **harmonic series**,

$$\sum_{k=1}^{\infty} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots$$

which arises in connection with the overtones produced by a vibrating musical string. It is not immediately evident that this series diverges. However, the divergence will become apparent when we examine the partial sums in detail. Because the terms in the series are all positive, the partial sums

$$s_1 = 1, \quad s_2 = 1 + \frac{1}{2}, \quad s_3 = 1 + \frac{1}{2} + \frac{1}{3}, \quad s_4 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}, \dots$$

form a strictly increasing sequence

$$s_1 < s_2 < s_3 < \cdots < s_n < \cdots$$

(Figure 9.3.4a). Thus, by Theorem 9.2.3 we can prove divergence by demonstrating that there is no constant M that is greater than or equal to *every* partial sum. To this end, we will consider some selected partial sums, namely, $s_2, s_4, s_8, s_{16}, s_{32}, \dots$. Note that the subscripts are successive powers of 2, so that these are the partial sums of the form s_{2^n} (Figure 9.3.4b). These partial sums satisfy the inequalities

$$s_2 = 1 + \frac{1}{2} > \frac{1}{2} + \frac{1}{2} = \frac{2}{2}$$

$$s_4 = s_2 + \frac{1}{3} + \frac{1}{4} > s_2 + \left(\frac{1}{4} + \frac{1}{4} \right) = s_2 + \frac{1}{2} > \frac{3}{2}$$

$$s_8 = s_4 + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} > s_4 + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} \right) = s_4 + \frac{1}{2} > \frac{4}{2}$$

$$\begin{aligned} s_{16} &= s_8 + \frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12} + \frac{1}{13} + \frac{1}{14} + \frac{1}{15} + \frac{1}{16} \\ &> s_8 + \left(\frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} \right) = s_8 + \frac{1}{2} > \frac{5}{2} \end{aligned}$$

$\vdots$

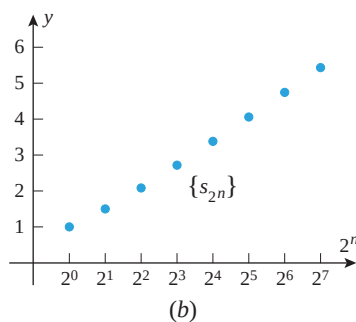
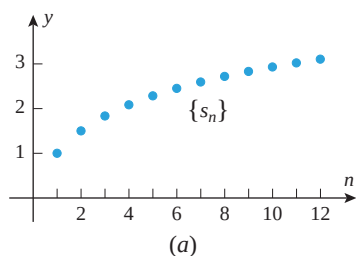
$$s_{2^n} > \frac{n+1}{2}$$

If M is any constant, we can find a positive integer n such that $(n+1)/2 > M$. But for this n

$$s_{2^n} > \frac{n+1}{2} > M$$

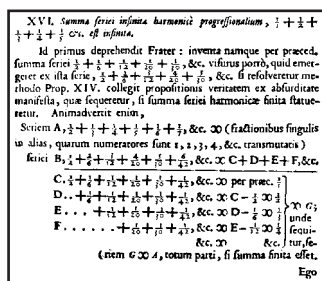
so that no constant M is greater than or equal to *every* partial sum of the harmonic series. This proves divergence.

This divergence proof, which predates the discovery of calculus, is due to a French bishop and teacher, Nicole Oresme (1323–1382). This series eventually attracted the interest of Johann and Jakob Bernoulli (p. 700) and led them to begin thinking about the general concept of convergence, which was a new idea at that time.



Partial sums for the harmonic series

▲ Figure 9.3.4



Courtesy Lilly Library, Indiana University
This is a proof of the divergence of the harmonic series, as it appeared in an appendix of Jakob Bernoulli's posthumous publication, *Ars Conjectandi*, which appeared in 1713.

 QUICK CHECK EXERCISES 9.3 (See page 623 for answers.)

1. In mathematics, the terms “sequence” and “series” have different meanings: a _____ is a succession, whereas a _____ is a sum.

2. Consider the series

$$\sum_{k=1}^{\infty} \frac{1}{2^k}$$

If $\{s_n\}$ is the sequence of partial sums for this series, then

$$s_1 = \text{_____}, s_2 = \text{_____}, s_3 = \text{_____},$$

$$s_4 = \text{_____}, \text{ and } s_n = \text{_____}.$$

3. What does it mean to say that a series $\sum u_k$ converges?

4. A geometric series is a series of the form

$$\sum_{k=0}^{\infty} \text{_____}$$

This series converges to _____ if _____. This series diverges if _____.

5. The harmonic series has the form

$$\sum_{k=1}^{\infty} \text{_____}$$

Does the harmonic series converge or diverge?

EXERCISE SET 9.3 CAS

1–2 In each part, find exact values for the first four partial sums, find a closed form for the n th partial sum, and determine whether the series converges by calculating the limit of the n th partial sum. If the series converges, then state its sum. ■

1. (a) $2 + \frac{2}{5} + \frac{2}{5^2} + \cdots + \frac{2}{5^{k-1}} + \cdots$

(b) $\frac{1}{4} + \frac{2}{4} + \frac{2^2}{4} + \cdots + \frac{2^{k-1}}{4} + \cdots$

(c) $\frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \frac{1}{4 \cdot 5} + \cdots + \frac{1}{(k+1)(k+2)} + \cdots$

2. (a) $\sum_{k=1}^{\infty} \left(\frac{1}{4}\right)^k$ (b) $\sum_{k=1}^{\infty} 4^{k-1}$ (c) $\sum_{k=1}^{\infty} \left(\frac{1}{k+3} - \frac{1}{k+4}\right)$

3–14 Determine whether the series converges, and if so find its sum. ■

3. $\sum_{k=1}^{\infty} \left(-\frac{3}{4}\right)^{k-1}$

4. $\sum_{k=1}^{\infty} \left(\frac{2}{3}\right)^{k+2}$

5. $\sum_{k=1}^{\infty} (-1)^{k-1} \frac{7}{6^{k-1}}$

6. $\sum_{k=1}^{\infty} \left(-\frac{3}{2}\right)^{k+1}$

7. $\sum_{k=1}^{\infty} \frac{1}{(k+2)(k+3)}$

8. $\sum_{k=1}^{\infty} \left(\frac{1}{2^k} - \frac{1}{2^{k+1}}\right)$

9. $\sum_{k=1}^{\infty} \frac{1}{9k^2 + 3k - 2}$

10. $\sum_{k=2}^{\infty} \frac{1}{k^2 - 1}$

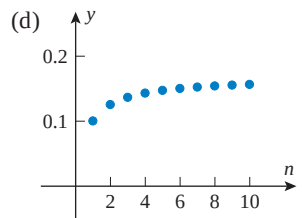
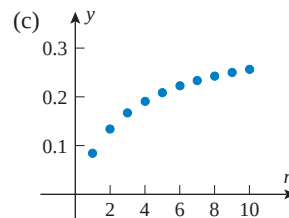
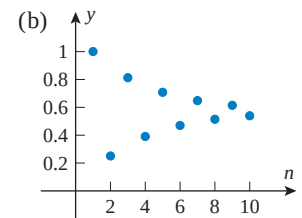
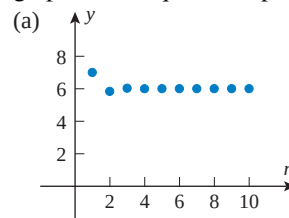
11. $\sum_{k=3}^{\infty} \frac{1}{k-2}$

12. $\sum_{k=5}^{\infty} \left(\frac{e}{\pi}\right)^{k-1}$

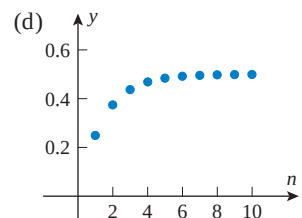
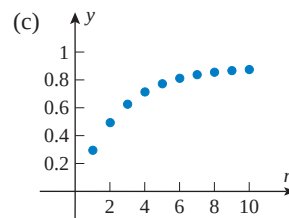
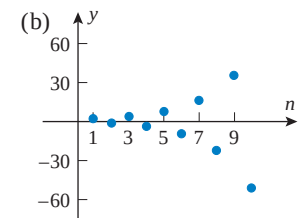
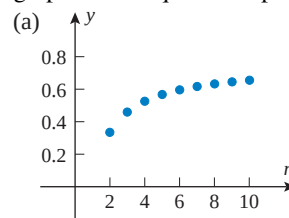
13. $\sum_{k=1}^{\infty} \frac{4^{k+2}}{7^{k-1}}$

14. $\sum_{k=1}^{\infty} 5^{3k} 7^{1-k}$

- 15.** Match a series from one of Exercises 3, 5, 7, or 9 with the graph of its sequence of partial sums.



- 16.** Match a series from one of Exercises 4, 6, 8, or 10 with the graph of its sequence of partial sums.



17–20 True–False Determine whether the statement is true or false. Explain your answer. ■

17. An infinite series converges if its sequence of terms converges.
18. The geometric series $a + ar + ar^2 + \cdots + ar^n + \cdots$ converges provided $|r| < 1$.
19. The harmonic series diverges.
20. An infinite series converges if its sequence of partial sums is bounded and monotone.

21–24 Express the repeating decimal as a fraction. ■

21. $0.9999\ldots$ 22. $0.4444\ldots$
23. $5.373737\ldots$ 24. $0.451141414\ldots$
25. Recall that a *terminating decimal* is a decimal whose digits are all 0 from some point on ($0.5 = 0.50000\ldots$, for example). Show that a decimal of the form $0.a_1a_2\ldots a_n9999\ldots$, where $a_n \neq 9$, can be expressed as a terminating decimal.

FOCUS ON CONCEPTS

26. The great Swiss mathematician Leonhard Euler (biography on p. 3) sometimes reached incorrect conclusions in his pioneering work on infinite series. For example, Euler deduced that

$$\frac{1}{2} = 1 - 1 + 1 - 1 + \cdots$$

and

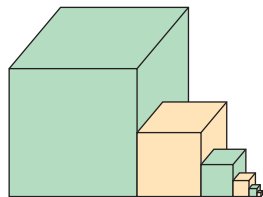
$$-1 = 1 + 2 + 4 + 8 + \cdots$$

by substituting $x = -1$ and $x = 2$ in the formula

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$$

What was the problem with his reasoning?

27. A ball is dropped from a height of 10 m. Each time it strikes the ground it bounces vertically to a height that is $\frac{3}{4}$ of the preceding height. Find the total distance the ball will travel if it is assumed to bounce infinitely often.
28. The accompanying figure shows an “infinite staircase” constructed from cubes. Find the total volume of the staircase, given that the largest cube has a side of length 1 and each successive cube has a side whose length is half that of the preceding cube.



◀ Figure Ex-28

29. In each part, find a closed form for the n th partial sum of the series, and determine whether the series converges. If so, find its sum.

(a) $\ln \frac{1}{2} + \ln \frac{2}{3} + \ln \frac{3}{4} + \cdots + \ln \frac{k}{k+1} + \cdots$

(b) $\ln \left(1 - \frac{1}{4}\right) + \ln \left(1 - \frac{1}{9}\right) + \ln \left(1 - \frac{1}{16}\right) + \cdots$
 $\quad \quad \quad + \ln \left(1 - \frac{1}{(k+1)^2}\right) + \cdots$

30. Use geometric series to show that

(a) $\sum_{k=0}^{\infty} (-1)^k x^k = \frac{1}{1+x}$ if $-1 < x < 1$

(b) $\sum_{k=0}^{\infty} (x-3)^k = \frac{1}{4-x}$ if $2 < x < 4$

(c) $\sum_{k=0}^{\infty} (-1)^k x^{2k} = \frac{1}{1+x^2}$ if $-1 < x < 1$.

31. In each part, find all values of x for which the series converges, and find the sum of the series for those values of x .

(a) $x - x^3 + x^5 - x^7 + x^9 - \cdots$

(b) $\frac{1}{x^2} + \frac{2}{x^3} + \frac{4}{x^4} + \frac{8}{x^5} + \frac{16}{x^6} + \cdots$

(c) $e^{-x} + e^{-2x} + e^{-3x} + e^{-4x} + e^{-5x} + \cdots$

32. Show that for all real values of x

$$\sin x - \frac{1}{2} \sin^2 x + \frac{1}{4} \sin^3 x - \frac{1}{8} \sin^4 x + \cdots = \frac{2 \sin x}{2 + \sin x}$$

33. Let a_1 be any real number, and let $\{a_n\}$ be the sequence defined recursively by

$$a_{n+1} = \frac{1}{2}(a_n + 1)$$

Make a conjecture about the limit of the sequence, and confirm your conjecture by expressing a_n in terms of a_1 and taking the limit.

34. Show: $\sum_{k=1}^{\infty} \frac{\sqrt{k+1} - \sqrt{k}}{\sqrt{k^2 + k}} = 1$.

35. Show: $\sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+2} \right) = \frac{3}{2}$.

36. Show: $\frac{1}{1 \cdot 3} + \frac{1}{2 \cdot 4} + \frac{1}{3 \cdot 5} + \cdots = \frac{3}{4}$.

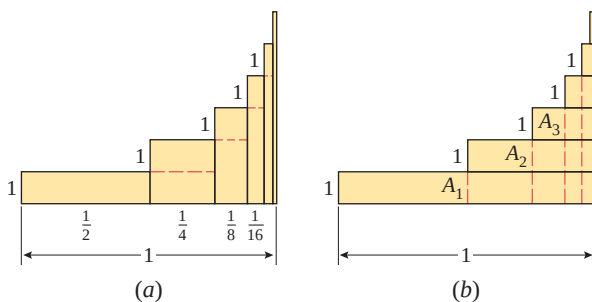
37. Show: $\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \cdots = \frac{1}{2}$.

38. In his *Treatise on the Configurations of Qualities and Motions* (written in the 1350s), the French Bishop of Lisieux, Nicole Oresme, used a geometric method to find the sum of the series

$$\sum_{k=1}^{\infty} \frac{k}{2^k} = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \cdots$$

In part (a) of the accompanying figure, each term in the series is represented by the area of a rectangle, and in

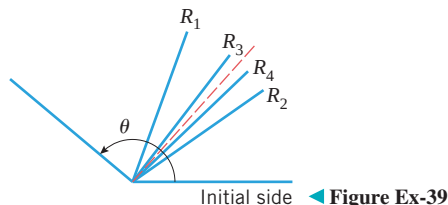
part (b) the configuration in part (a) has been divided into rectangles with areas $A_1, A_2, A_3, \dots$. Find the sum $A_1 + A_2 + A_3 + \dots$.



▲ Figure Ex-38

39. As shown in the accompanying figure, suppose that an angle θ is bisected using a straightedge and compass to produce ray R_1 , then the angle between R_1 and the initial side is bisected to produce ray R_2 . Thereafter, rays $R_3, R_4, R_5, \dots$ are constructed in succession by bisecting the angle between the preceding two rays. Show that the sequence of angles that these rays make with the initial side has a limit of $\theta/3$.

Source: This problem is based on “Trisection of an Angle in an Infinite Number of Steps” by Eric Kincannon, which appeared in *The College Mathematics Journal*, Vol. 21, No. 5, November 1990.



◀ Figure Ex-39

40. In each part, use a CAS to find the sum of the series if it converges, and then confirm the result by hand calculation.

(a) $\sum_{k=1}^{\infty} (-1)^{k+1} 2^k 3^{2-k}$ (b) $\sum_{k=1}^{\infty} \frac{3^{3k}}{5^{k-1}}$ (c) $\sum_{k=1}^{\infty} \frac{1}{4k^2 - 1}$

41. **Writing** Discuss the similarities and differences between what it means for a sequence to converge and what it means for a series to converge.
42. **Writing** Read about Zeno’s dichotomy paradox in an appropriate reference work and relate the paradox in a setting that is familiar to you. Discuss a connection between the paradox and geometric series.

✓ QUICK CHECK ANSWERS 9.3

1. sequence; series 2. $\frac{1}{2}; \frac{3}{4}; \frac{7}{8}; \frac{15}{16}; 1 - \frac{1}{2^n}$ 3. The sequence of partial sums converges.
4. ar^k ($a \neq 0$); $\frac{a}{1-r}$; $|r| < 1$; $|r| \geq 1$ 5. $\frac{1}{k}$; diverge

9.4 CONVERGENCE TESTS

In the last section we showed how to find the sum of a series by finding a closed form for the n th partial sum and taking its limit. However, it is relatively rare that one can find a closed form for the n th partial sum of a series, so alternative methods are needed for finding the sum of a series. One possibility is to prove that the series converges, and then to approximate the sum by a partial sum with sufficiently many terms to achieve the desired degree of accuracy. In this section we will develop various tests that can be used to determine whether a given series converges or diverges.

■ THE DIVERGENCE TEST

In stating general results about convergence or divergence of series, it is convenient to use the notation $\sum u_k$ as a generic notation for a series, thus avoiding the issue of whether the sum begins with $k = 0$ or $k = 1$ or some other value. Indeed, we will see shortly that the starting index value is irrelevant to the issue of convergence. The k th term in an infinite series $\sum u_k$ is called the **general term** of the series. The following theorem establishes